

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

## Practical - III

|                               |                                   |
|-------------------------------|-----------------------------------|
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| Class :SYB.SC.IT              | Batch : B1                        |
| Date of Assignment :02/7/2026 | Date/Time of Submission :027/2026 |

Stack Application:

Write a program to:

- a. Implement a stack using an array.

Code:

```
import array as arr
```

```
stack1 = arr.array('i')  
si = int(input("Enter size of stack: "))  
print(stack1)
```

```
def push():  
    print("Push Operation")  
    if len(stack1) == si:  
        print("Stack Overflow")  
    else:  
        ele = int(input("Enter Element: "))  
        stack1.append(ele)  
        print(stack1)
```

```
def pop():  
    print("Pop Operation")  
    if isEmpty() == 0:  
        print("Stack Underflow")  
    else:  
        stack1.pop()  
        print(stack1)
```

```
def peek():
    print("Peek Operation")
    if isEmpty() == 0:
        print("Stack is Empty")
    else:
        print("Top Element =", stack1[-1])
```

```
def isEmpty():
    if len(stack1) == 0:
        return 0
    else:
        return 1
```

```
num = 0
```

```
while num != 5:
    print("\n1.Push")
    print("2.Pop")
    print("3.Peek")
    print("4.isEmpty")
    print("5.Exit")
```

```
num = int(input("Select Choice: "))
```

```
if num == 1:
    push()
```

```
elif num == 2:
    pop()
```

```
elif num == 3:
    peek()
```

```
elif num == 4:
    if isEmpty() == 0:
        print("Stack is Empty")
    else:
        print("Stack is Not Empty")
```

```
elif num == 5:
    print("PROGRAM CLOSED")
```

```
else:
    print("Invalid Choice")

print("S056 Shweta Singh")
```

## Output:-

```
=====
IDLE Shell - C:/Users/Admin/OneDrive - illusion Dental Laboratory DivisionofLaxmi Dental/Documents/dfgyu.py (3.14.6)
File Edit Shell Debug Options Window Help
=====
Enter size of stack: 2
array('i', [0])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 1
Push Operation
Enter Element: 24
array('i', [24])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 1
Push Operation
Enter Element: 13
array('i', [24, 13])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 1
Push Operation
Stack Overflow

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 3
Peek Operation
Top Element = 13

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 3
Peek Operation
Top Element = 13

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Peek Operation
array('i', [24])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Peek Operation
array('i', [24])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Pop Operation
array('i', [24])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Pop Operation
array('i', [24])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Pop Operation
Stack Underflow

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Pop Operation
Stack Underflow

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 4
Stack is Empty
array('i', [0])

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Pop Operation
Stack Underflow

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 4
Stack is Empty

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 2
Pop Operation
Stack Underflow

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 4
Stack is Empty

1. Push
2. Pop
3. Peek
4. isEmpty
5. Exit
Select Choice: 5
PROGRAM CLOSED
S056 Shweta Singh
>>>
```

#### Curiosity Questions:

1. What's the difference between `peek()` and `pop()` in stack operations?
2. Why do we call a stack a "LIFO" structure? Can you think of a real-life example of this?
3. Can we store more than one data type (like `int` and `string`) in a stack using array module? Why or why not?
4. What happens if we try to `peek()` into an empty stack?
5. How can we check if the stack is empty before using `pop()`?